Question

Adding a slightly excessive amount of concentrated oxalic acid solution to a solution of nitrate $\mathbf{M}(NO_3)_3$ produces no gas. The precipitated precipitate is washed with an ethanol-water mixed solvent to obtain white crystals $\mathbf{A}$. The decomposition products of $\mathbf{A}$ are not completely the same under different atmospheres.

Thermogravimetric analysis of $\mathbf{A}$ in an argon atmosphere shows a weight loss of 20.9% when heated from $90°C$ to $300°C$, and a further weight loss of 23.5% when heated to $500°C$. The valence of $\mathbf{M}$ does not change in this step; further heating to $800°C$ results in a further weight loss of 4.1%, and the residual solid is a black powder, the composition of which is a mixture of the oxide $\mathbf{B}$ $(\mathbf{MO_2})$ and another substance.

Heating $\mathbf{A}$ in air to $450°C$ results in a weight loss of 49.7%, and the only remaining solid is oxide $\mathbf{B}$.

The following statements are correct:

A. All other options are incorrect

B. $\mathbf{M}$ ground state has a single electron

C. The relative molecular mass of $\mathbf{A}$ is less than 660.

D. The black powder does not contain any elemental substance.

E. $\mathbf{A}$ The second step of weight loss in an argon atmosphere only produces one kind of gas.

F. $\mathbf{A}$ The solid remaining after the second-step weight loss in an argon atmosphere is a pure substance.

G. $\mathbf{A}$ In the solid remaining after the second step of weight loss in an argon atmosphere, the chemical formula of the M-containing substance contains 8 atoms.

Answer

Correct Answer: G

Detailed Explanation

The addition of a slightly excessive amount of concentrated oxalic acid solution to a solution of nitrate $\mathbf{M}(NO_3)_3$ produces no gas, so no redox reaction occurs;

CHECKPOINT 1 PTS

Precipitate formation without redox reaction

The precipitate formed should be hydrated oxalate, and the valence of $\mathbf{M}$ should still be +3, so $\mathbf{A}$ can be written as $\mathbf{M}_2(C_2O_4)_3 \cdot nH_2O$.

CHECKPOINT 1 PTS

The precipitate formed should be hydrated oxalate, and $\mathbf{A}$ can be written as $\mathbf{M}_2(C_2O_4)_3 \cdot nH_2O$

$\mathbf{M}_2(C_2O_4)_3 \cdot nH_2O$ decomposes in air, producing only oxide $\mathbf{M}O_2$ and losing 49.7% of its weight, so the following equation can be written:

$$\frac{x+32}{x+264.1+18n} = 1 - 0.497$$

$\mathbf{M}_2(C_2O_4)_3 \cdot nH_2O$ loses weight in argon, and the first step is likely to be the loss of water of crystallization; therefore, the following equation is written:

CHECKPOINT 1 PTS

$\mathbf{M}_2(C_2O_4)_3 \cdot nH_2O$ loses weight in argon, and the first step is likely to be the loss of water of crystallization

$\frac{18n}{x+264.1+18n} = 0.209$

The simultaneous equations can be solved to get $n = 8.07 \approx 8$. The solution is not an integer because of errors in the thermogravimetric analysis.

CHECKPOINT 1 PTS

Simultaneous equations can be solved to get the number of water of crystallization $n = 8.07 \approx 8$

Substituting $n = 8$, the solution is $x = 143.6$. Near the atomic weight of 144, considering that $\mathbf{M}$ has a stable +4 valence oxide, it should be Ce. The electronic configuration of Ce is $4f^1 5d^1 6s^2$, with two unpaired electrons, so option B is incorrect.

CHECKPOINT 1 PTS

Considering that $\mathbf{M}$ has a stable +4 valence oxide, it should be Ce

Therefore, the chemical formula of $\mathbf{A}$ is $Ce_2(C_2O_4)_3 \cdot 8H_2O$. The relative molecular mass is 688.4, so option C is incorrect.

CHECKPOINT 1 PTS

$\mathbf{A}$ chemical formula is $Ce_2(C_2O_4)_3 \cdot 8H_2O$

The amount of weight loss in the second step of $Ce_2(C_2O_4)_3 \cdot 8H_2O$ in argon is $688.4 \times 0.235 = 162$, which should be $3CO_2 + CO$ (the difference in molecular weight is 2, which is the difference between the theoretical and actual weight loss). Since CO is produced, a redox reaction should have occurred. Since the valence of Ce has not changed, only elemental carbon can be produced.

CHECKPOINT 1 PTS

The amount of weight loss in the second step of $Ce_2(C_2O_4)_3 \cdot 8H_2O$ in argon is 162, which should be $3CO_2 + CO$

CHECKPOINT 1 PTS

The valence of Ce has not changed, so only elemental carbon can be produced

Therefore, the equation can be balanced:

$$Ce_2(C_2O_4)_3 = 3CO_2 + CO + Ce_2O_2(CO_3) + 2C$$

CHECKPOINT 1 PTS

$$Ce_2(C_2O_4)_3 = 3CO_2 + CO + Ce_2O_2(CO_3) + 2C$$

Therefore, the solid produced in the second step is $\mathrm{Ce_2O_2(CO_3) + 2C}$, and the black solid in the third step is $\mathrm{CeO_2 + C}$.

CHECKPOINT 1 PTS

The solid produced in the second step is $\mathrm{Ce_2O_2(CO_3) + 2C}$, and the black solid in the third step is $\mathrm{CeO_2 + C}$

Therefore, options D, E, and F are all incorrect, and option G is correct.